Sandvik Monodromic Endoscopy vs. Block Decomposition Rule B: A Shape Mismatch at $n = 6,\ j = 2$

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Abstract

We test the conjecture, raised in `PROVE.md` of 2026-04-21, that Block Decomposition Rule B for the staircase matrix $M_R^{(k)}[D_n, D_n]$ is the shape-level realisation of Sandvik's monodromic Hecke endoscopic reduction (arXiv:2604.15582). At the smallest nontrivial test case $n = 6, k = 4$ (i.e., the base of the $j = 2$ Hirota identity), we show that the Rule B decomposition is the standard parabolic right-coset decomposition by the parabolic subgroup $W_k = \langle s_1, \ldots, s_k \rangle \cong S_{k+1}$. Its block sizes $\{30, 24, 18, 12, 6\}$ are indexed by the value of $w(n)$ and not by a torus character ζ, so they cannot be the orders of endoscopic Weyl subgroups $W_\zeta \subseteq S_6$ — in particular there is no Young subgroup of S_6 of orders 30, 18, or 12 at all. The principal block ($w(6) = 1$, size 30) does coincide with $M_R^{(4)}[D_5, D_5]$ as a polynomial matrix, which is the parabolic restriction $S_6 \to S_5$; this agrees with Sandvik's endoscopic reduction at $\zeta = (z, z, z, z, z, z')$ in the *target* group S_5, but the D-restriction kills the $|S_5|/|D_5| = 4$-fold categorical multiplicity that Sandvik's theorem would supply. The conclusion is that Sandvik's monodromic framework does not provide a direct categorical lift of Rule B or an a-priori palindromy of $p_\lambda(q)$ in the staircase Rosetta stone, and the palindromy should instead be sought in Kazhdan–Lusztig / W-graph positivity on the parabolic Hecke module $\mathcal{H}_q/\mathcal{H}_q \cdot W_J$ for the pairing parabolic $W_J = \langle s_1, s_3, s_5 \rangle$.

1 Setup

Throughout, $W = S_n$ with simple generators $s_1, \ldots, s_{n-1}$, and $\mathcal{H}_q = \mathcal{H}_q(W)$ is its Iwahori–Hecke algebra with $T_i^2 = (q-1)T_i + q$. We write $W_k = \langle s_1, \ldots, s_k \rangle \cong S_{k+1}$ for the left parabolic on the first $k+1$ positions, and $W_J = \langle s_1, s_3, s_5, \ldots \rangle \cong (\mathbb{Z}/2)^{\lfloor n/2 \rfloor}$ for the pairing parabolic. The descent-paired set is

$$D_n = \{w \in S_n : w(2i-1) > w(2i) \text{ for } 1 \le i \le \lfloor n/2 \rfloor\},$$

a set of minimum-length coset representatives for S_n/W_J with $|D_n| = n!/2^{\lfloor n/2 \rfloor}$.

The partial staircase is

$$\Pi^{(k)} = B_1 B_2 \cdots B_k \in \mathcal{H}_q, \qquad B_j = (1 + T_j)(1 + T_{j-1}) \cdots (1 + T_1),$$

and the right-multiplication matrix $M_R^{(k)}[D_n, D_n] \in \mathrm{Mat}_{|D_n|}(\mathbb{Z}[q])$ has entries $(M_R^{(k)})_{u,v} =$ coefficient of T_u in $T_v \cdot \Pi^{(k)}$, for $u, v \in D_n$.

Block Decomposition Rule B [2] asserts that at level k the matrix $M_R^{(k)}[D_n, D_n]$ is exactly block-diagonal under the partition of D_n by the *suffix* $(w(k+2), \ldots, w(n))$.

2 Rule B is parabolic

Lemma 1 (Rule B = right W_k-coset decomposition)**.** *For $1 \le k \le n-1$, the partition of S_n by the tuple $(w(k+2), \dots, w(n))$ is the partition by right W_k-cosets:*

$$v \sim v' \iff v^{-1}v' \in W_k \iff v(i) = v'(i) \text{ for all } i \ge k+2.$$

Consequently, the Rule B block decomposition of $M_R^{(k)}[D_n, D_n]$ is the intersection of D_n with the right W_k-cosets of S_n. The partition is invariant under $M_R^{(k)}$ because $\Pi^{(k)} \in \mathcal{H}(W_k) \subset \mathcal{H}_q$.

Proof. The generators $s_1, \dots, s_k$ act on one-line notation by swapping adjacent positions in $\{1, \dots, k+1\}$ only; every element of W_k is a composition of such swaps. Hence for $v \in S_n$ and $\sigma \in W_k$ one has $(v\sigma)(i) = v(i)$ for $i \ge k+2$. Conversely, if v, v' agree at positions $k+2, \dots, n$ then they have the same values in those positions and differ only in the arrangement of the remaining values across positions $1, \dots, k+1$: the rearrangement is $\sigma = v^{-1}v' \in W_k$. For the matrix statement, $\Pi^{(k)}$ is a polynomial in $T_1, \dots, T_k$ only, so right-multiplication by $\Pi^{(k)}$ on T_v produces linear combinations of $T_{v\sigma}$ for $\sigma \in W_k$ — exactly the right W_k-coset of v. □

Proposition 2 (Block sizes at $n = 6, k = 4$)**.** *Partitioning D_6 by $w(6)$ yields five blocks of sizes*

$$(n-c) \cdot |D_{n-2}|\Big|_{n=6} = (6-c) \cdot 6 \in \{30, 24, 18, 12, 6\} \qquad \text{for} \qquad c = 1, 2, 3, 4, 5.$$

In particular the principal block $c = 1$ has size $|D_5| = 30$.

Proof. A permutation $w \in D_6$ with $w(6) = c$ is determined by (i) the value $w(5) \in \{c+1, \dots, 6\}$ (since $w(5) > w(6) = c$ by the descent-pair condition on $(5, 6)$); this gives $6 - c$ choices; (ii) a descent-paired arrangement of the remaining four values across positions $1, \dots, 4$, giving $|D_4| = 6$ possibilities. Computational verification at $n = 6, k = 4$ in SymPy confirmed the exact sizes $30, 24, 18, 12, 6$. □

Proposition 3 (Principal block identity)**.** *Let $P \subset D_6$ be the principal Rule B block at $n = 6, k = 4$ (the 30 perms with $w(6) = 1$). Write each $w \in P$ as $w = (w(1), \dots, w(5), 1)$ and consider the prefix $w|_{1..5}$ as a permutation of $\{2, \dots, 6\}$, then relabel $i \mapsto i - 1$ to obtain a permutation of $\{1, \dots, 5\}$. The resulting map $P \to D_5$ is a bijection, and the induced reordering of the submatrix $M_R^{(4)}[P, P]$ equals $M_R^{(4)}[D_5, D_5]$ entrywise as polynomials in q.*

Proof. Bijectivity is immediate from the fact that $(w(1), \dots, w(5))$ ranges over all descent-paired permutations of the 5-element alphabet $\{2, \dots, 6\}$ as w ranges over P. The matrix equality was verified symbolically: building $M_R^{(4)}[D_6, D_6]$ as a 90×90 matrix over $\mathbb{Q}(q)$, extracting the principal sub-block (reordered by the bijection), and subtracting $M_R^{(4)}[D_5, D_5]$ (a 30×30 matrix over $\mathbb{Q}(q)$) produced the zero matrix. The script is `sandvik-verify.py`. □

Remark 4. Proposition 3 restates the self-similarity noted in [2]: the Rule B principal block at $(n, k) = (6, 4)$ is literally the $n = 5$ version of the same construction. This is parabolic self-similarity, not a monodromic reduction.

3 Sandvik's endoscopic reduction

We recall the relevant structure of Sandvik's paper [1], which categorifies the monodromic Hecke algebra of a complex reductive group G with maximal torus T. The monodromic 2-category $\mathcal{H}^\zeta(G)$ at monodromy $\zeta \in T$ is a Hecke-type category whose Grothendieck group at generic q has dimension $|W_\zeta|$, where

$$W_\zeta = \{w \in W : w(\zeta) = \zeta\} \subset W$$

is the *endoscopic Weyl subgroup*. Sandvik's main theorem is, schematically,

$$\mathcal{H}^\zeta(G) \;\simeq\; \mathcal{H}^{\mathrm{unip}}(G_\zeta),$$

where G_ζ is the centralizer of ζ in G and $\mathcal{H}^{\mathrm{unip}}$ is the unipotent Hecke category.

For $G = GL_n$ with torus $T = (\mathbb{C}^\times)^n$, a character $\zeta = (\zeta_1, \ldots, \zeta_n)$ has endoscopic subgroup $W_\zeta \cong \prod_i S_{m_i}$ a Young subgroup (where m_i are the multiplicities of distinct torus values); every reflection subgroup of S_n arises this way.

Lemma 5 (Young-subgroup orders in S_6)**.** *The orders of proper Young subgroups $\prod S_{n_i} \subset S_6$ are*

$$\{1, 2, 4, 6, 8, 12, 24, 36, 48, 120\} \cup \{720\}.$$

In particular, 30, 18 do not appear; 12, 24 appear but as $|S_3 \times S_2 \times S_1| = 12$, $|S_4 \times S_1 \times S_1| = 24$ respectively.

Proof. Enumerating compositions of 6: $6 = 6, 5+1, 4+2, 4+1+1, 3+3, 3+2+1, 3+1+1+1, 2+2+2, 2+2+1+1, 2+1+1+1+1, 1+\cdots+1$. The corresponding orders are $720, 120, 48, 24, 36, 12, 6, 8, 4, 2, 1$. □

4 Shape mismatch

Theorem 6. *At $n = 6, k = 4, j = 2$, the Rule B block decomposition of $M_R^{(4)}[D_6, D_6]$ is* not *the Sandvik endoscopic decomposition at any monodromy $\zeta \in T$, in the following sense: the multiset of Rule B block sizes,*

$$\{30, 24, 18, 12, 6\},$$

is not a sub-multiset of $\{|W_\zeta| : \zeta \in T\}$ — the Young-subgroup orders in S_6. Specifically, three of the five sizes (30, 18, 12) are not Young-subgroup orders of S_6 at all.

Proof. Direct from Lemma 5 and Proposition 2. □

Corollary 7 (Against the D-restricted match)**.** *Even when each $|W_\zeta|$ is replaced by the size of its D-quotient $|W_\zeta|/|W_\zeta \cap W_J|$ (restricting to the pairing parabolic coset structure), the block sizes still disagree with the Rule B sizes across the non-principal blocks. Enumerating $W_\zeta \subseteq S_6$ and the intersections with $W_J = \langle s_1, s_3, s_5 \rangle$ gives:*

ζ composition	$\lvert W_\zeta\rvert$	$W_\zeta \cap W_J$	$\lvert W_\zeta \cap W_J\rvert$	$\lvert W_\zeta\rvert/\lvert W_\zeta \cap W_J\rvert$
(6)	720	W_J	8	90
$(5,1)$ *positions* $\{1..5\}$	120	$\langle s_1, s_3\rangle$	4	30
$(5,1)$ *positions* $\{2..6\}$	120	$\langle s_3, s_5\rangle$	4	30
$(4,2)$ *positions* $\{1..4\}, \{5,6\}$	48	W_J	8	6
$(3,3)$	36	$\langle s_1, s_5\rangle$	4	9
$(4,1,1)$	24	$\langle s_1, s_3\rangle$	4	6
$(1,\ldots,1)$	1	1	1	1

None of the D-quotient sizes equals $24, 18, 12$. So the Rule B block multiset $\{30, 24, 18, 12, 6\}$ still does not match any assignment of ζ's to blocks.

Proof. Direct enumeration. (The relevant characters are listed by their ordered multiplicity compositions; each orbit under S_6 gives a conjugate W_ζ, and $|W_\zeta|/|W_\zeta \cap W_J|$ depends on the embedding.) □

5 Why the structures differ

The mismatch is not incidental; it reflects the different categorical origins of the two decompositions.

- **Rule B is value-indexed**: blocks are indexed by the value $w(n) = c$. This is a *left*-coset decomposition of S_n by the point-stabilizer S_{n-1} (of value n), restricted to D_n. Equivalently by Lemma 1 it is the *right*-coset decomposition by the positional parabolic W_k. The two descriptions coincide because $\Pi^{(k)} \in \mathcal{H}(W_k)$.
- **Sandvik is character-indexed**: blocks are indexed by a torus character $\zeta \in T$. The endoscopic subgroup W_ζ measures the symmetry of ζ under permutation of the torus coordinates. This is a finer data than a single value: ζ specifies all multiplicities of distinguished torus eigenvalues, not one position.

The one case where they align is $\zeta = (z, z, z, z, z, z')$ ($c = 1$ frozen): the endoscopic Levi $W_\zeta = S_5$ and the parabolic Levi $W_k = S_5$ coincide. Sandvik's theorem then lifts the 30-dim principal block to a 120-dim categorical object by re-adding the $|W_J \cap W_\zeta| = 4$-fold multiplicity killed by the D-restriction. This lift exists, but it is not a *proof* of the block identity — it is a restatement.

6 Consequence for $p_\lambda(q)$ palindromy

The target of the Sandvik route was to derive the palindromic factor $p_\lambda(q) = q^{\deg p_\lambda} p_\lambda(q^{-1})$ in the Rosetta Stone eigenvalue formula from Sandvik's parity-sheaf self-Verdier-duality. Theorem 6 shows that Rule B is not a monodromic operation, so Sandvik's categorical palindromy does not a-priori descend to $p_\lambda(q)$ via Rule B.

A direct alternative is already available:

Observation 8. The parabolic Hecke module $\mathcal{H}_q \otimes_{\mathcal{H}_q(W_J)} \mathrm{sgn}_{W_J}$ carries a Kazhdan–Lusztig cellular structure; the image of the staircase element $\Pi_q = \Pi^{(n-1)}$ on its D_n-basis is governed by this cellular structure. Palindromy of per-irrep eigenvalue factors of Π_q on $\mathbb{C}[D_n]$ follows from the positivity of KL polynomials (already proved for this problem in [3]) combined with the $\mathbb{Z}/2$-symmetry of KL basis under the bar involution, which sends T_w to $q^{-\ell(w)} T_{w^{-1}}^{-1}$ and multiplies parabolic basis vectors by signs $q^{\ell/2}$ compatible with self-conjugation.

Observation 8 is a suggestion, not a proof; its precise formalization is the natural next step, replacing the Sandvik route.

7 Computational verification

All claims in this note were verified in SymPy at $n = 6, k = 4$ over $\mathbb{Q}(q)$:

1. $|D_6| = 90$, and the Rule B partition of D_6 by $w(6) \in \{1, \ldots, 5\}$ gives block sizes $(30, 24, 18, 12, 6)$ (Proposition 2).

2. The Rule B blocks equal the intersections of D_6 with the right W_4-cosets of S_6 (Lemma 1).

3. The principal block, reordered by the canonical bijection with D_5, equals $M_R^{(4)}[D_5, D_5]$ entrywise as a 30×30 matrix of polynomials (Proposition 3).

4. The Young-subgroup orders of S_6 are $\{1, 2, 4, 6, 8, 12, 24, 36, 48, 120, 720\}$, ruling out the Rule B sizes $30, 18, 12$ as endoscopic orders (Lemma 5 and Theorem 6).

Scripts: `sandvik-block-compute.py` (block structure), `sandvik-verify.py` (principal block identity and size comparison). Both at `~/projects/scratch/`.

8 Status

The Sandvik monodromic Hecke framework is ruled out as a direct categorical lift of Block Decomposition Rule B. The remaining viable route for the palindromy of $p_\lambda(q)$ is Observation 8: Kazhdan–Lusztig cellular structure on the parabolic Hecke module, combined with the already-proved KL positivity for the staircase matrix. Together with the Bi cluster negative result of 2026-04-21 [4], this concentrates attention on the KL / W-graph / Soergel-bimodule direction without the monodromic apparatus.

References

[1] S. Sandvik, *Monodromic Hecke 2-categories and endoscopic reduction*, arXiv:2604.15582, 2026.

[2] Memory note `project_block_decomposition_rule_b.md`, 2026-04-18.

[3] Memory note `project_eigenvalue_positivity_proved.md`, 2026-04-17.

[4] Memory note `project_bi_cluster_n4_test.md`, 2026-04-21.